

1 Introduction

1. Definition of masting
 - (a) Observations of masting in relation to seed predation and flowering
2. Different hypotheses operating at different reproductive stages like filters
 - (a) First, pollination success (Pollination coupling)
 - (b) Second, resources availability during development phase (Resource Matching)
 - (c) Third, dispersal (Predation satiation)
3. Why traits can help link hypotheses to masting mechanisms
 - (a) Mechanisms are not mutually exclusive, but some might act stronger than the other for certain species
 - (b) Traits thus can be important to understand masting through their relative importance
4. Trait-based perspective
 - (a) Literature review on understanding masting from traits perspective
 - i. Wind pollination is related to masting
 - ii. Animal dispersed seeds have higher masting than wind dispersed seeds
 - iii. Larger seed is associate with masting
 - iv. All these studies use CV to quantify masting, explain the pros and cons of CV.
 - v. If evolutionary associations are sufficiently strong, we should be able to detect it even under a coarse binary classification
 - vi. Recent papers also focus on resource perspective using leaf and shoot traits
 - (b) Existing studies mainly focus on ecological mechanisms.
 - (c) Next step may be clear connection between traits and hypotheses and potential evolutionary drivers for certain species
5. Introduce framework linking hypothesis x traits x masting
 - (a) This framework is especially useful when considering whether evolutionary drivers are more similar within closely related species (clades)
 - (b) Linking traits to specific hypotheses helps us understand the underlying drivers of masting across different species
6. We try to do this using data from one specific geographical region

- (a) Here we use North American tree data to test this framework since they have information on their life-history
 - i. Masting info available
 - ii. Includes both strong masting and weak masting species
 - iii. We try to understand this on species level, so we are using a binary trait here instead of CV

2 Discussion

1. Summary of main findings
 - (a) Pollination supports the pollination coupling hypothesis
 - (b) Seed weight supports the predation satiation hypothesis
 - (c) These act at the beginning and end of the reproductive cycle, two critical stages for successful reproduction and gene transmission
 - (d) Discuss mechanisms in the context of existing literature
 - i. Previous literature on seed mass and masting think from the resource depletion perspective (Bigger seeds use up resources so cannot reproduce every year)
 - ii. However, from evolutionary perspective, seeds attractive to animal dispersers will benefit more from masting
 - iii. At the same time, they have to make sure the seed survived can established
 - iv. Selection may favor coordinated strategies that increase seed survival across the whole recruitment process (Bonal *et al.* (2007); Moles & Westoby (2004))
 - v. Relationship between masting and seed weight is stronger within biotic-dispersed group
 - (e) Other traits show some pattern but statistically non-significant
 - (f) Transition to alternative hypotheses
2. Masting is a population-level phenomenon
 - (a) Other traits may matter to masting below the species level, like population level
 - (b) Examples of species that mast only under certain environmental conditions (E.g., *Hymenaea coubaril*)
 - (c) To address, this traits need to be measured locally and match with masting data – we need more data below the level of species!
3. Different species selected under different hypotheses
 - (a) Introduce flowering masting vs. seed masting

- (b) Different species driven by different mechanisms
 - (c) Need a more refined definition of masting and additional flowering-related traits
 - (d) Transition to phylogeny
4. Phylogenetic signal for traits and masting
 - (a) Many traits have strong phylogenetic signal, it is hard to tease out traits from masting, which limits the inference
 - (b) The lit on this mostly talk about adaptive significance, we need to understand its broader macroevolutionary patterns
 - (c) Data is strongly nested in phylogeny, and data is sparse.
 5. Deep evolutionary split between angiosperm and gymnosperm
 - (a) Gymnosperms and angiosperms occupy fundamentally different functional spaces
 - i. More masting gymnosperm sp
 - ii. Longer leaf longevity in gymnosperm
 - iii. More monoecious species in gymnosperm
 - iv. Wind pollination is universal among gymnosperms
 - v. Importance of looking for potential evolution support between gymnosperm and angiosperm
 - A. Masting is more common in gymnosperm with specific functional traits
 - B. Test if masting was gradually diluted in angiosperm as they transitioned to biotic pollination and faster life histories
 6. Importance of testing within clade
 - (a) Existing literature points out the potentials within certain clade for evolutionary investigations (Satake & Kelly (2021))
 - (b) This requires more data collection on species-level and population-level within certain genus groups, for example: (*Quercus*.)
 - (c) So we need more data lower than species and more targeted data across species that considers deeper evolutionary time (we're missing both the super recent evolution. – populations – and the super deep)

References

- Bonal, R., Munoz, A. & Díaz, M. (2007) Satiation of predispersal seed predators: the importance of considering both plant and seed levels. *Evolutionary Ecology* **21**, 367–380.

- Moles, A.T. & Westoby, M. (2004) Seedling survival and seed size: a synthesis of the literature. *Journal of Ecology* **92**, 372–383.
- Satake, A. & Kelly, D. (2021) Studying the genetic basis of masting. *Philosophical Transactions of the Royal Society B: Biological Sciences* **376**.